

Strengthening Communities: Towards AI-Empowered Knowledge Management for Community Social Services

Oghenemaro Anuyah

University of Notre Dame, Department of Computer Science and Engineering
Notre Dame, Indiana, United States
oanuyah@nd.edu

ABSTRACT

The welfare and well-being of vulnerable and marginalized individuals, such as those facing poverty and food or housing insecurity, heavily rely on their access to essential social services. These services are commonly provided by community social service organizations, which are known to deal with multifaceted needs and must collaborate with a network of other social services. Operating in such distributed networks presents unique challenges in knowledge management and knowledge transfer (KM/KT), which are critical for effectively serving these populations. My dissertation research focuses on studying KM/KT within these distributed social services networks. My dissertation comprises four studies that: 1) Examine technology integration barriers and opportunities in community social services. 2) Explore knowledge management processes within these organizations to identify how technology can better support their work. 3) Investigate the potential of AI in community social services through participatory design. and 4) Implement and evaluate a technological intervention for community service knowledge management.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**;
Field studies; **Interactive systems and tools**.

KEYWORDS

Marginalized Communities, Community-Based Participatory Research, Social Services, Knowledge Management, Knowledge Sharing, Responsible Artificial Intelligence

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1 BACKGROUND AND MOTIVATION

The HCI community has extensively studied the intersection of vulnerable and marginalized communities and technology, resulting in significant contributions across various domains—from health

and wellness [19] to safety [22], and the overarching challenges of technological integration [5]. These studies not only demonstrate the transformative potential of technology in empowering communities [9, 18] but also highlight the technological challenges faced by marginalized groups due to social, cultural, and systemic factors [1, 11, 16]. While HCI research has increasingly focused on direct interventions for individuals within vulnerable and marginalized populations, the role of community-based organizations as intermediaries in this process has not been as thoroughly examined. Studies engaging with community-based organizations to improve services like job skills and violence prevention demonstrate the value of collaborative technology design [7, 8]. However, the specific challenges of how these community organizations capture, manage, and maintain information and knowledge and the role of technological interventions in this context remain underexplored in the literature.

The significance of addressing the unique challenges in the social service sector becomes more evident when considering the distributed nature of service providers, who typically operate in high-demand and low-resource settings. These community-based organizations, including non-profits, local private entities, and religious institutions [6, 12, 13, 17, 23], often face considerable resource constraints while serving vulnerable and marginalized populations. These organizations not only experience high turnover rates but also struggle with the complex task of meeting the multifaceted needs of those they support. Populations in need of their services frequently require a range of assistance, extending beyond the scope of a single organization. For example, individuals requiring basic needs like feeding assistance and clothing may also need mental health support, housing, and transportation services [3]. This necessitates collaboration and information and knowledge sharing across various organizations. The ad-hoc practices typically used for knowledge management and transfer within these community organizations are often inadequate to meet the complex demands of this interconnected and distributed network.

Building on previous HCI research, my dissertation research focuses on leveraging intelligent systems and semi-automation to enhance knowledge capture and sharing within distributed community organizations. The potential transformative impact of emerging AI capabilities, particularly Large Language Models (LLMs), is increasingly evident across various domains within HCI [15, 21]. This impact is highlighted by recent studies, such as the use of LLMs for nutrition assistance in low-income communities [21], showcasing the diverse and practical applications of these technologies. Further studies have also demonstrated how LLMs can enhance processes and refine communication in various contexts [14, 15],

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thereby proving their versatility and ability to optimize workflows and improve interactions.

Despite these advancements, there remains a significant unexplored potential for using LLMs for knowledge management within community-based social service organizations. This gap presents a promising avenue of research, particularly in exploring human-AI collaboration in the context of community social services. Investigating this aspect could lead to significant improvements in how these organizations manage, share, and utilize knowledge, potentially transforming the effectiveness of the services they provide and their impact on the community.

By exploring the design and implementation of collaborative systems between human users and AI technologies, my dissertation aims to address complex, real-world challenges in knowledge management, particularly in constrained resources. My work will contribute to the field of HCI by developing a technology intervention specifically designed to empower marginalized and vulnerable communities. My dissertation research seeks to demonstrate how intelligent systems can be ethically and effectively utilized to enhance the capacity of social service organizations, bridging gaps in knowledge management and fostering greater community support.

2 RESEARCH OBJECTIVES

The objectives of my doctoral research are developed to meet the overarching goal of enhancing knowledge management and knowledge transfer in community social service contexts through the integration of AI and semi-automation. With this in mind, the primary research questions guiding my dissertation work are as follows:

- (1) What are the technological barriers within community social service organizations, and what opportunities exist to enhance access to and effectiveness of social services?
- (2) How do community service organizations manage, maintain, and capture information and knowledge, and how do these processes compare with those in large corporations? How can technology improve these knowledge management processes, given strategies adopted in large corporate environments?
- (3) How can participatory design and AI be leveraged to develop and evaluate effective, responsible technological solutions for knowledge management in the distributed environments of community social service organizations?

To achieve these objectives, I use a combination of community-based participatory research methods, such as ethnographic fieldwork, semi-structured interviews, and participatory design studies, to collect data. Additionally, I utilize thematic analysis to interpret and synthesize the collected data. Throughout this research, I leverage my partnerships with four community organizations in Northern Indiana, USA. This community currently faces high levels of poverty in several of its neighborhoods, leading to an increased demand for social services over the years. The challenging socioeconomic landscape of this community makes it a suitable location for my dissertation research.

3 THEORETICAL FRAMEWORK

To help guide our research and design partnerships with community service organizations, we adopted Maslow's hierarchy of needs to develop a conceptual framework published in CHI 2023 [2]. This framework is tailored to assist HCI researchers in identifying and addressing the technology-related needs of vulnerable and marginalized communities, thereby enabling more empowering and effective collaborative efforts. Central to this framework—and our research—is the imperative of engaging with marginalized communities thoughtfully and empathetically, fully acknowledging their unique challenges. It is critical to our work because it is common for those serving marginalized and vulnerable groups to share similar backgrounds or experiences [7].

To create this framework, I synthesized findings from 84 research articles, focusing on how various levels of Maslow's hierarchy—from basic to self-actualization needs—are addressed in technology settings for vulnerable groups. Through a thematic analysis of these articles, I identified specific technology-related needs across different levels of Maslow's hierarchy. I categorized these technology needs into two types: deficits-based and assets-based, drawing upon Maslow's theory of motivation. Deficits-based needs include fundamental requirements like access to information and communication technologies, digital literacy skills, and the need for safety and privacy. Assets-based needs are derived from the inherent strengths of these populations, such as their social ties and identities and the capacity to contribute their perspectives to research and design. I incorporated theories of intersectionality and design justice to explore dimensions pertinent to vulnerable populations, thereby providing a comprehensive understanding of their technology needs. Utilizing empowerment concepts by Schneider et al. [20], I illustrated how meeting these needs could lead to empowering outcomes for vulnerable and marginalized communities, ultimately supporting their active participation in research and design.

Throughout my research, I have diligently applied the guiding principles of our framework in my interactions and partnerships with community organizations, particularly focusing on addressing issues like technology access and digital literacy. In addition to avoiding extractive data collection practices, I actively contribute to our partner organizations by volunteering my time to assist with various operational needs, including tasks like handling and distributing donations, serving in their kitchens, and tackling computer-related challenges such as data entry.

4 PRIOR AND PROPOSED RESEARCH APPROACH, METHODS, AND OUTPUTS

This section provides an overview of my four dissertation studies.

4.1 Study 1: An Interview Study Investigating Technology Integration Opportunities and Barriers

This study was published in the ACM Journal of Computing and Sustainable Societies and presented at COMPASS 2023 [3]. To explore the intersection of technology and community-based social service provision, I conducted semi-structured interviews with 21 representatives from various community organizations in Northern Indiana,

USA. These organizations serve a population in a city marked by high demand for social services and constrained resources.

Through affinity diagramming analysis of the interview data, it became evident that these organizations struggle with information and knowledge management, a situation further exacerbated by the heightened service demands during the COVID-19 pandemic. The interviews highlighted key technological barriers such as infrastructure inadequacies within community organizations and challenges in digital literacy among community service workers (CSWs),¹ while also revealing the resourcefulness and existing strengths of these organizations. This understanding is critical in informing the design and development of practical technological solutions. The contributions of this work to the literature were twofold: 1) We provided an empirical understanding of how technology can enhance access to social services, considering both the organizations and the populations seeking services, and 2) We offered a comprehensive set of design recommendations for researchers aiming to empower organizations that provide social services and the communities they serve.

4.2 Study 2: An Ethnography of Knowledge Management Processes and Challenges

In my second study, I conducted an in-depth ethnographic investigation within two community social service organizations in Northern Indiana, USA, focusing on their knowledge management processes and challenges, particularly in the context of housing insecurity. My immersion into these organizations over a period of three months was complemented by seven interviews with CSWs from both organizations. I conducted twenty-one on-site visits to our partner organizations and observed a total of eight staff and inter-organizational meetings. These interactions were instrumental in providing insights into how these organizations share, maintain, and capture knowledge and information, and the challenges they face in carrying out these processes.

Findings from the thematic analysis [4] of field notes and interview transcripts revealed how CSWs navigate their dynamic work environments using everyday communication and collaboration tools (emails and Google Docs). These tools were essential for their immediate capture and subsequent dissemination of information. However, despite their willingness to adopt advanced technologies to improve their work, significant barriers to technological adoption and coordination across complex needs were identified as key challenges. Our research identified parallels in knowledge management practices and challenges between these community organizations and large corporations, particularly in their shared emphasis on fostering critical organizational learning for their workers, cultivating a culture of continuous knowledge sharing, and the necessity of developing communities of practice within these organizations to effectively overcome coordination and knowledge sharing challenges. Our contributions to the HCI literature from this study are multifaceted: 1) We provide an in-depth understanding of the processes employed by CSWs to manage, share, and capture information and knowledge. 2) We introduce a human-social-technical

framework, conceptualizing the essential components of knowledge management in HCI and highlighting the interplay between human factors, technology, and socio-organizational contexts. 3) We offer a novel comparison of knowledge management processes between large corporate organizations and small community service organizations. 4) We propose design implications for enhancing collaboration and knowledge sharing in community social services aimed at empowering these vital organizations.

4.3 Study 3: Participatory Design Workshops Examining the Future of Community Social Services with AI

For my third study, I aim to build upon the insights gained from my second study (the ethnography) by utilizing speculative futuring and scenario-based research methodologies. This approach is designed to explore the potential roles of AI within community social services, drawing inspiration from scenario-based research methods and design workbooks as introduced by Harrington et al. [10]. The study will proceed in a two-phase approach. The first phase involves identifying and analyzing tasks within community social services suitable for AI collaboration. The second phase will engage CSWs in participatory design workshops to co-create and refine scenarios illustrating the potential integration of AI into service provision. Participants will actively contribute to the design of future AI applications, assessing both the practical benefits and challenges and ensuring that the technological solutions are aligned with the real-world needs of community service organizations. CSWs will actively be research and design collaborators on our research team throughout this research. We will utilize our key contacts within our partner organizations to help recruit 10–12 CSWs for these workshops. In addition to my research role, I will maintain my commitment to adding value to these research partnerships by volunteering at their organization.

An important outcome of this study will be a methodological contribution to conducting AI research with communities. This includes demystifying the research process to ensure community members can effectively engage with AI and contribute to design and idea generation. Additionally, in collaboration with community members, we will explore the potential concerns and limitations of AI. Moreover, this study will yield a set of design recommendations based on the outputs of the workshops. These recommendations will focus on the implications of the workshop-generated designs in developing solutions for knowledge management in community social services. These design recommendations will be instrumental in informing how AI can be responsibly and effectively integrated into the work of community social services, ensuring that the solutions are both technologically robust and socially beneficial. The goal is to contribute to responsible AI practices that are mindful of the social and ethical implications of technology in community settings.

¹We use the term “Community Service Workers” to refer to staff or volunteers who directly interact with and serve guests (or community residents in need) in these community organizations.

4.4 Study 4: Design and Evaluation of an AI-Empowered Technology Intervention for Knowledge Management in Community Social Services

Based on the insights from Study 3, I will develop a technology intervention aimed at enhancing knowledge management practices within community social service organizations. This intervention will incorporate an AI model tailored to facilitate human-AI collaboration in the distributed environments of community social service organizations. The goal is to enable these organizations to effectively share, capture, and receive updates on information and knowledge to facilitate the effective provision of services to the individuals and families who need them. Throughout this research, CSWs will participate as research and design collaborators.

The deployment of this solution in community organizations will be followed by an evaluation phase consisting of observations and a user study. Again, throughout the research, I will maintain a commitment to ensure a mutually beneficial partnership, such as through volunteering at the organization. This research will significantly contribute to HCI by demonstrating effective strategies for implementing AI solutions in organizations with limited access to and utilization of AI and advanced technology. It will establish a model for validating AI approaches in contexts that benefit from adaptable and incremental technological advancements.

5 NEXT STEPS, DISCUSSION, CHALLENGES

The first two studies of my dissertation have been successfully completed. These studies have established a solid foundation for understanding the technological landscape and knowledge management practices and challenges within community social service contexts. As of Spring 2024, I am actively progressing with the third study. Following IRB approval, I have started planning the workshops and recruiting CSW participants. These workshops are scheduled to begin in mid-spring. The fourth study is planned to begin in the Summer of 2024. This stage will focus on implementing, deploying, and evaluating technology interventions for knowledge management and knowledge transfer with partner community service organizations.

The anticipated contributions of my dissertation work to HCI are multifaceted: 1) Providing an empirical understanding of technology integration opportunities and barriers within community social service organizations, thereby enhancing access to social services and offering design recommendations. 2) Conducting an in-depth exploration of knowledge management processes and challenges in these organizations, resulting in a novel Human-Socio-Technical Framework. 3) Advancing methodological approaches in AI research with communities, especially through participatory design workshops. 4) Developing and evaluating an AI-empowered technology intervention for knowledge management in community social services, demonstrating effective AI implementation strategies for environments that currently underutilize advanced technologies, and offering a practical model for adaptable AI integration.

Alongside these developments, several challenges and open questions remain in my research trajectory. A primary challenge involves integrating AI effectively within the unique contexts of

community social services, taking into account varying levels of technological infrastructure and digital literacy. The scalability of AI-driven solutions across different organizations and their adaptability to diverse community needs also pose significant open questions. Additionally, ensuring that these technological interventions align with the ethical standards and cultural sensitivities of the communities we serve is essential.

Another layer of challenge arises from the operational realities of our partner organizations. Their high demand and limited resources often leave little room for staff and volunteers to engage in research activities. To alleviate this, I have committed to volunteering and assisting with tasks such as handling donations and administrative support, thereby easing their workload. This approach not only aids these organizations operationally but also strengthens our research partnership, ensuring more meaningful and representative participation in our studies. While this approach is beneficial, it requires careful management to ensure that our research objectives are met without placing additional burdens on these organizations.

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